21.8 HOOP: A Scalable Hybrid DC-DC Converter Ring for High-Performance Computing

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In the AI era, demands of computing power grow exponentially, driving the need for power delivery solutions that offer *stronger* power capability, *higher* efficiency and *faster* response, *together* [1]. To satisfy the enormous current demand, tens or hundreds of point-of-load converters operate in parallel to multiply the output current capability. However, maintaining high efficiency in multi-phase converters requires dedicated current-balance control among each converter phase (Fig. 21.8.1). Traditionally, this is achieved by integrating a global digital controller, which monitors the inductor currents and adjusts the PWM signals to ensure current balancing. However, as the number of phases (N) increases, the complexity of maintaining current balance escalates, presenting additional challenges [2]. Hence, an inherent auto-current-balancing characteristic of the DC-DC converter is highly desirable. Prior arts [3-5] leverage the charge balance properties of capacitors to obtain automatic inductor current balancing. Similar approaches have been employed to balance inductor currents in conventional buck and three-level converters [6-8]. However, when these hybrid structures are used in parallel to increase output capability, ensuring the current balance between different converters remains a significant challenge. In this paper, we have designed a **H**ybrid converter with a power ring (**O**) for current balance and a back ring (**O**) for phase shedding for computing **P**ower delivery, as a **HOOP**.

The proposed converter features a ring-shaped structure around the xPU loads, facilitating automatic current balancing and easy extension to any number of converter phases. This approach involves extracting a small, fixed portion of the inductor current from each branch and passing it to the next branch, i.e. each branch also incorporates the same portion of current from the previous branch. As a result, if a current mismatch occurs between two adjacent inductors, the difference in the sink and source currents across the branches will automatically correct the imbalance. To further enhance efficiency across varying load currents, a phase shedding back ring is also incorporated. The extracted current from each cell can be fed into either the next cell or the first cell, ensuring efficient operation under different load conditions. This auto-balanced ring structure simplifies the power delivery layout for xPU applications. By placing the converter on the backside of the load in a ring configuration, the current can be delivered vertically to the xPUs via short paths, greatly reducing both the on-board and in-package I^2R losses [9].

To demonstrate the concept, this work presents the designed module with the proposed hybrid ring cells (HRCs). The HRC (Fig. 21.8.2) incorporates two flying capacitors and two inductors. Each HRC consists of two branches, with each branch operating in three distinct states. In phase α_X, inductor L_X charges flying capacitor C_{FX}. In phase β_X, inductor L_X discharges the preceding capacitor C_{FX-1}. In phase γ_X, inductor L_X is de-magnetized. Since C_{FX-1} is utilized in both the α_{X-1} and β_X phases, these two phases must be interleaved. The duty cycles of states α_X and β_X are set to the same duration, DT. By applying the voltage-second balance principle in the inductors and charge balance in the flying capacitors, we can conclude that the inductor currents automatically balance across different inductors and the flying capacitor voltages will equalize as well. Besides, despite the absence of a dedicated charging path to maintain a fixed voltage on the flying capacitors, a natural balancing loop ensures that the flying capacitor voltage stabilizes at $V_{IN}/2$ [10,11]. Figure 21.8.2 (bottom) illustrates an example of the proposed HOOP module with three HRCs. In this configuration, the three HRCs are connected in parallel, with the POST pin of each chip connected to the PRE pin of the subsequent chip. Additionally, all of the BACK pins are all connected to the PRE pin of the first chip. For phase shedding, the HRCs can be configured into two modes: active mode and sleep mode. In active mode operation, each HRC can either connect forward to the next one or return to the first HRC to form a complete current balance loop. In sleep mode, the HRC is inactive, enhancing efficiency at lower output currents. Moreover, to achieve fast transient response, a transient mode is also introduced. In this mode, all power branches operate in the α phase, simultaneously increasing the current in all inductors and thereby enhancing the load transient response performance [5].

Figure 21.8.3 (left) illustrates the block diagram of the designed HRC, which integrates all power transistors (M_1-M_7), along with their drivers, bootstrap circuits, and controller, into a single chip. Switches M_1 and M_2 are implemented using 6V LDMOS, while M_3-M_7 are implemented using 12V LDMOS. For switches M_1-M_5, a conventional cascaded bootstrap structure with an active diode implemented by PMOS can be used [12,13]. However, switch M_6 presents a challenge because its source terminal is connected to the POST pin, and the operating mode of the subsequent chip is not fixed. If the subsequent chip is in sleep mode for phase shedding, the bootstrap capacitor of M_6 cannot be charged, preventing M_6 from operating correctly. To address this issue, we use the bootstrap capacitor of M_7 to charge that of M_6. The bootstrap capacitor of M_7 is charged from that of M_3, based on the principle that V_{BACK} equals to $V_{IN}/2$ and SW_1 switches between $V_{IN}/2$ and 0. This approach ensures that all power switches operate correctly, without being affected by the phase shedding of other chips. Besides, due to the long distance between M_3 and M_7 in the layout, a discrete off-chip capacitor, C_{BOOT7}, is placed in parallel with the on-chip bootstrap capacitor. Moreover, an off-chip capacitor is also added to maintain the DC voltage of the BACK node and to address minor misalignment of the phases of the first and the last active chips.

Due to its inherent auto-current-balance property, the HOOP can be controlled simply using a primary-secondary voltage mode PID control, without sensing any inductor currents. The duty cycle is generated by the first HRC chip and send to each HRC. Furthermore, because adjacent converters operate out of phase, propagation delay is not a major concern. A mode transition protector is introduced to eliminate ultra-short-pulse signals during mode transitions and to protect the chip. For hybrid converters, transitioning a chip from sleep mode to active mode presents another challenge, the flying capacitor and bootstrap voltages must be maintained during sleep mode. If these voltages are not maintained, the power switches may experience overvoltage and fail to operate correctly when the chip is reactivated [14]. In this work, we address this issue by incorporating dedicated diodes D_{CLAMP} and M_{CLAMP}, along with parasitic diodes D_{BS7} and D_{M7}, to clamp the voltages of the flying capacitors and bootstrap capacitors, as shown in Fig. 21.8.3 (top right). According to the load current level and load transients, the HRC chips switch between four modes, following the rules given in Fig. 21.8.3 (bottom right).

Figure 21.8.4 (top left) shows the HRC start-up waveforms, demonstrating the auto current balance and flying capacitor voltage balance across the entire V_{IN} range. Figure 21.8.4 (top right) presents the steady-state waveforms of a single HRC chip at an I_{OUT} of 3A, achieving a current mismatch of only 40mA and an output voltage ripple of 18mV. Figure 21.8.4 (bottom left) shows the measured waveforms during phase shedding, with the second chip disabled or enabled. Figure 21.8.4 (bottom right) shows the detail of phase shedding process, highlighting the voltage leakage and the effect of the voltage clamp.

Figure 21.8.5 (top left) exhibits the current balance performance in the presence of inductor mismatch, demonstrating the tolerance of current balance against path impedance. Despite significant differences in inductance values, L_1 (2μH, 204mΩ) and L_2 (1μH, 48mΩ), the inductor currents exhibit only a 90mA mismatch. Figure 21.8.5 (top right) illustrates the load transient response with a load increase from 300mA to 3A within 80ns. With the HRCs operating in transient mode, not only do I_{L1} and I_{L2} rise together, but I_{L3} and I_{L4} of the sleep mode HRCs also help charge the output capacitor, accelerating the recovery of the output voltage and reducing the voltage droop by 50%. The implemented HOOP module achieves a peak efficiency of 90.2% at 1A for a single chip and a peak efficiency of 87.2% at 4.2A for four chips, with a maximum output current of 16A. Figure 21.8.5 (bottom right) presents the analysis of additional power loss in the ring. In the case of four HRCs operating at 12-to-1V with an output current of I_{OUT}, the current transmitted across the ring path is only $I_{OUT}/8$, with a duty cycle of 1/12. At the peak efficiency point of 4A, this loss accounts for just 0.06% of the output power. Figure 21.8.6 provides the comparison table of the HOOP module with state-of-the-art designs [15], highlighting that this work offers an easily-scalable, auto-current-balanced solution with the highest power density and significant current density.

As shown in Fig. 21.8.7, the proposed HRC chip, fabricated in the 0.18μm BCD process, occupies a die area of 4.56mm^2 with two 10μF (0402) flying capacitors soldered on the die. Each HRC occupies an area of 21.5mm^2 with two 1μH (0806) inductors on the top side, and other passive components on the back side. The four-chip HOOP module occupies an area of 16x16mm^2 with all components on the top side of the PCB and the load on the back side, demonstrating its suitability for vertical power delivery applications.

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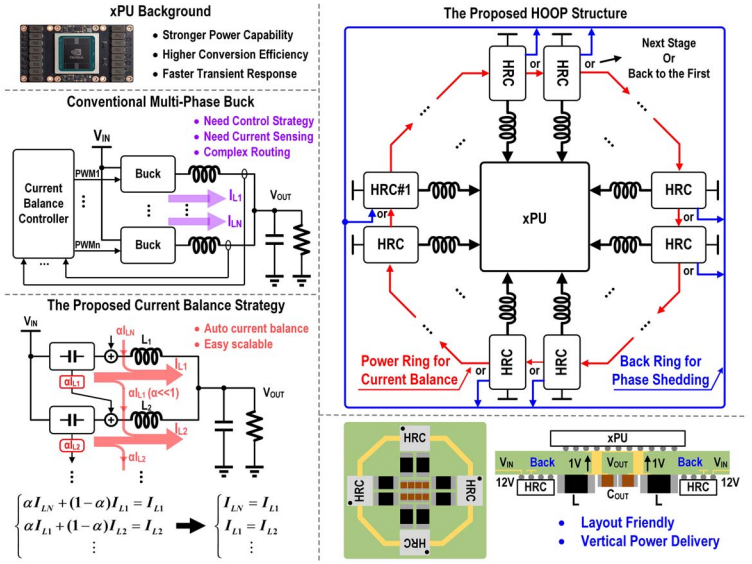


Figure 21.8.1: The multi-phase buck converter and the proposed current balance strategy featuring a HOOP structure.

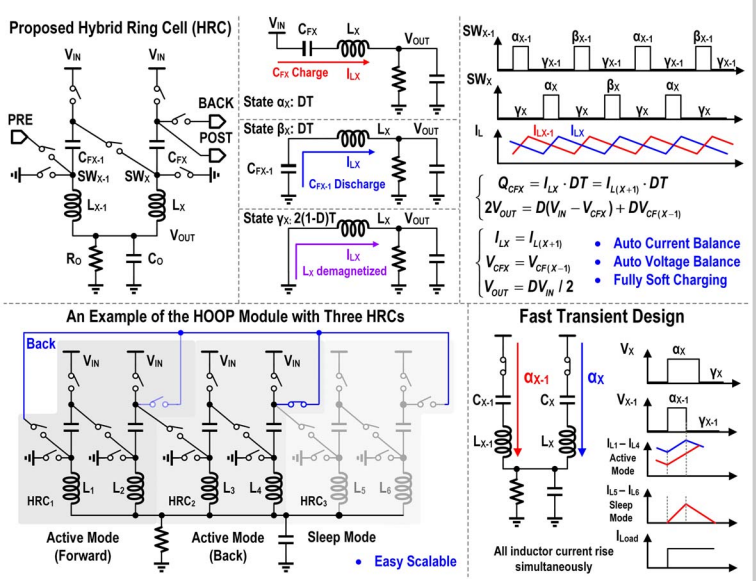


Figure 21.8.2: The proposed hybrid ring cell (HRC) and its operation principle.

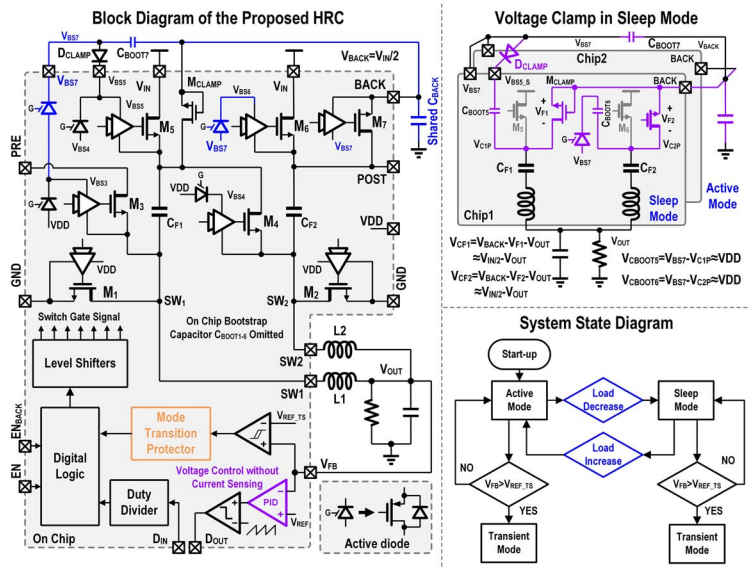


Figure 21.8.3: The block diagram of the proposed HRC chip and the circuit implementation.



Figure 21.8.4: The measured start-up, steady state, and phase shedding waveforms.

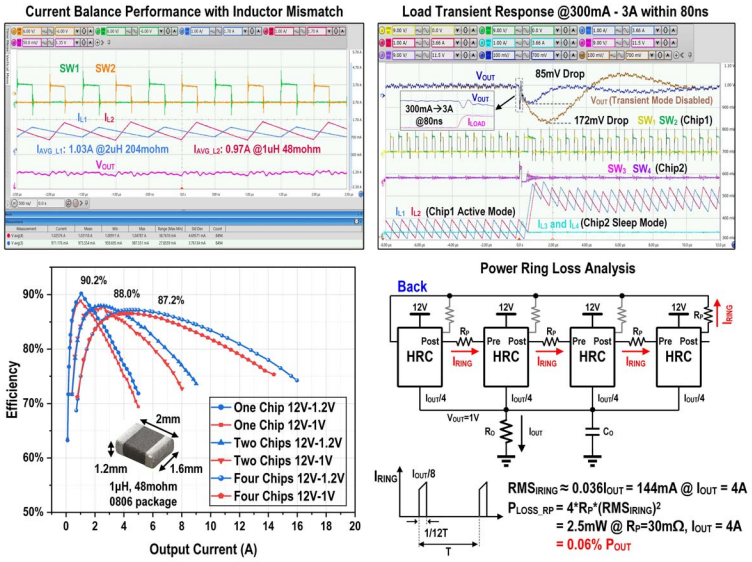
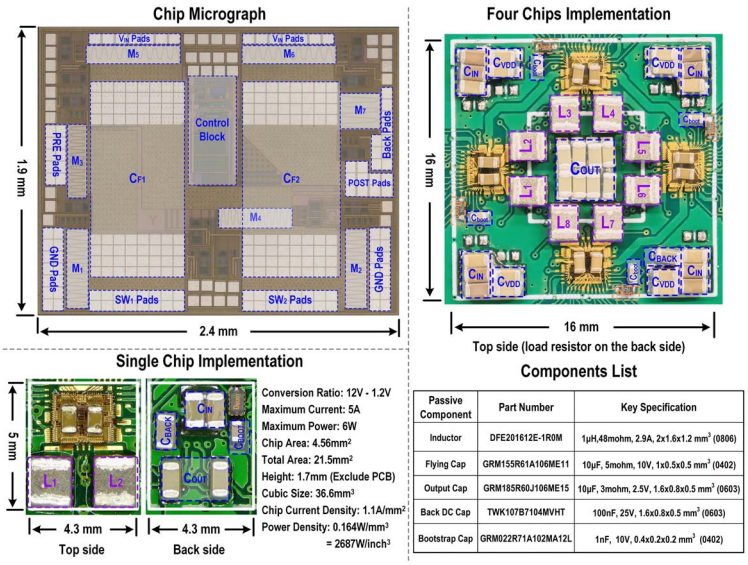


Figure 21.8.5: The measured current balance, load transient response waveforms, efficiency and loss analysis.

	TPE' 21[6]	ISSCC'22 [2]	JSSC'23 [4]	ADI MAX77542 [13]	CICC'24 [12]	This Work	
Process	65nm	180nm BCD	180nm SOI	N. A.	180nm BCD	180nm BCD	
Topology	2P3L Buck	DSD	CCC	MP-Buck	MP-CCC	HOOP	
Current balance Strategy	Auto-balance	Auto-balance	Auto-balance	Sense + Control	Sense + Control	Auto-balance Ring	
Scalable	NO	NO	NO	YES	YES	YES	
Control Scheme	Voltage Mode	Voltage Mode	Current Mode	Current Mode	Current Mode	Voltage Mode	
Norminal Input Voltage	1.8V	12V	12V	12V	12V	12V	
Norminal Output Voltage	0.3 - 1.6 V	1-1.2V	0.9 - 1.8V	0.3 - 5.2 V	1.2 - 1.8 V	1 - 1.2 V	
Switching Frequency	40MHz	1MHz	2MHz	0.5 - 1.5MHz	2MHz	1MHz	
Inductor L (DCR)	500nH (1.2mohm)	1.8μH x 2 (N.A.)	0.74μH x 2 (14.5mohm)	0.47μH x 4 (17mohm)	1μH x 4 (30mohm)	1μH x 2 (48mohm)	
Flying Capacitor	2.4nF x 2	4.7μF x 2	2.2μF x 2	N. A.	4.7μF x 4	10μF x 2	
Output Capacitor	1nF	10μF	10.6μF	22μF x 24	4.7μF x 4	10μF x 2	
Chip/Package Area	2.9mm^2	9.6mm^2	2.8mm^2	11.2mm^2	6.4mm^2	4.6mm^2	
Chip Current Density (A/mm^2)	0.07	0.42	1.43	1.4	0.63	1.1	
Number of Scalable Chips	N. A.	N. A.	N. A.	1	1	1	4
Peak Efficiency (output current) @12-1.2V	N. A.	89.4% (2A)	86.8% (1.8A)	85% (0.9A)	90.3% (1.6A)	90.2% (1.05A)	87.2% (4.2A)
Maximum Output Current	210mA	4A	4A	16	4	5A	16A
Power Density (W/inch^3)	N. A.	N. A.	294	1119	847	2687	1025

Figure 21.8.6: The comparison table with the state of art works.

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Passive Component	Part Number	Key Specification
Inductor	DFE201612E-1R0M	1μH,48mohm, 2.9A, 2x1.6x1.2 mm³ (0806)
Flying Cap	GRM155R61A106ME11	10μF, 5mohm, 10V, 1x0.5x0.5 mm³ (0402)
Output Cap	GRM185R60J106ME15	10μF, 3mohm, 2.5V, 1.6x0.8x0.5 mm³ (0603)
Back DC Cap	TWK107B7104MVHT	100nF, 25V, 1.6x0.8x0.5 mm³ (0603)
Bootstrap Cap	GRM022R71A102MA12L	1nF, 10V, 0.4x0.2x0.2 mm³ (0402)

Figure 21.8.7: The chip micrograph, PCB implementation and components list.

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